

Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters

1. Introduction

Afficionado Coffee Roasters operates across multiple store locations with a diverse menu of coffee, tea, and food products. Despite having access to transaction-level sales data, the business lacks structured product analytics, leaving menu decisions largely intuition-driven. This analysis addresses three core gaps: visibility into product popularity versus revenue contribution, category-level revenue dependency, and identification of underperforming menu items.

2. Dataset Description

Size: The dataset contains 149,116 rows and 11 columns covering transaction-level sales data from 2025 across three store locations.

Columns:

- **transaction_id:** Unique identifier per transaction
- **transaction_qty:** Quantity of units purchased per transaction
- **unit_price:** Price per unit of the product
- **store_id:** Unique identifier for each store
- **store_location:** Physical location of the store
- **product_id:** Unique identifier for each product
- **product_category:** Broad product group (e.g. Coffee, Tea, Bakery)
- **product_type:** Product variant within the category (e.g. Brewed Chai Tea, Barista Espresso)
- **product_detail:** Detailed product attributes such as flavor or blend
- **year:** Transaction year
- **transaction_time:** Time of transaction

Tools & Libraries:

- Python (pandas, matplotlib, duckdb) — DuckDB used to run SQL queries directly on dataframes
- Streamlit — used to build the interactive product performance dashboard
- Jupyter Notebook — used for data cleaning, normalization, and exploratory analysis

3. Methodology

3.1 Data Cleaning

Column names were standardized to lowercase with spaces replaced by underscores for consistency across queries and code. Date columns were converted from string to datetime format. No missing values or duplicate records were found in the dataset.

3.2 Normalization

The dataset was normalized into three tables — Transactions, Products, and Stores. Product attributes (product_category, product_type, product_detail) were extracted into a Products table as they are properties of the product itself rather than the transaction. Store location was isolated into a Stores table since it is a stable attribute of the store. The Transactions table retains all transaction-specific fields and references both tables via product_id and store_id as foreign keys. After normalization, the Products table contains 80 unique products and the Stores table contains 3 stores.

3.3 Feature Engineering

Three features were derived from the existing data. A revenue column was calculated by multiplying transaction_qty by unit_price. Volume rank and revenue rank were then assigned to each product using dense ranking — volume rank based on total units sold and revenue rank based on total revenue generated. These rankings enabled direct comparison between a product's popularity and its financial contribution, forming the basis of the popularity versus revenue analysis.

4. Exploratory Data Analysis

4.1 Product Popularity Analysis

Popularity was measured using total units sold across all transactions. Coffee emerged as the most popular product category, accounting for the highest units sold, followed by Tea as a distant second. At the product type level, Brewed Chai Tea and Gourmet Brewed Coffee led in units sold, while Green Beans recorded the lowest. Analysis was limited to the top 15 product types for clarity given the breadth of the menu.

4.2 Revenue Analysis

Revenue was examined at both the category and product type level. Coffee and Tea collectively accounted for 67% of total revenue, with Coffee alone contributing 39% and Tea 28%, indicating a significant financial dependency on these two categories. At the product type level, Barista Espresso generated the highest revenue followed by Brewed Chai Tea and Hot Chocolate, while Green Beans contributed the least. Product-level revenue was further examined using average revenue per product as a benchmark; products falling significantly below this benchmark were identified as low contributors relative to the broader menu.

4.3 Pareto Analysis

An 80/20 analysis was conducted on product-level revenue to identify revenue anchors and long-tail products. The analysis revealed that approximately the top 40 products out of 80 account for 80% of total revenue, with Products 61 and 59 serving as the strongest individual revenue anchors. This indicates that revenue is relatively evenly distributed across the menu compared to a typical Pareto distribution, where 20% of products would be expected to drive 80% of revenue. The remaining 40 products form a long tail of minimal contributors, each generating relatively little revenue individually.

4.4 Popularity vs Revenue

A correlation analysis between volume rank and revenue rank returned a moderate positive correlation of 0.61, meaning products that sell more units tend to generate more revenue but this is not always the case. A scatter plot with a fitted regression line confirmed this relationship visually, showing that while there is a general upward trend, several products diverge, selling high volumes at low revenue or generating strong revenue despite lower sales volumes. These divergences represent the most actionable insights for menu optimization decisions.

5. Limitations

The dataset was clean and well structured, leaving few analytical limitations. The primary constraint was the absence of customer-level data such as ratings, reviews, or purchase history, which prevented a deeper assessment of menu diversification risk. Additionally, with only one year of transaction data available, trend analysis over time could not be conducted.

6. Conclusion

This analysis delivered structured product performance insights across four areas: product popularity, revenue distribution, menu concentration, and the relationship between popularity and revenue. Coffee and Tea collectively account for 67% of total revenue, indicating a significant category dependency. The moderate correlation of 0.61 between popularity and revenue suggests that high sales volume does not always translate to strong revenue contribution, pointing to opportunities in menu pricing and product mix optimization. Additional data such as customer ratings and delivery dates would be needed to extend the analysis further.